process human language. Their "large" designation refers to the immense scale of the data they are trained on (billions of words, sentences, and documents) and the vast number of parameters they contain, allowing them to learn complex patterns in language. Essentially, LLMs work by predicting the most probable next word or sequence of words in a given context, enabling them to produce human-like text, answer questions, summarize information, and more. Tools like ChatGPT rely on these powerful LLMs to perform their "magic" by transforming user prompts into coherent and relevant creative outputs.

UIs and conversational components

The user interface (UI) is what users interact with when using the web application with a browser. It includes HTML elements such as buttons, forms, and menus, providing a seamless experience for user queries and receiving AI-generated outputs. Beyond traditional browser-based UIs, generative AI also powers conversational AI components such as chatbots, virtual assistants, and sophisticated AI agents. These agents are programs designed to perform specific tasks autonomously, often by interacting with various other AI APIs and external systems and services, enabling AI functionalities even within nonbrowser applications or operating in the background.

Backend infrastructure

Behind the scenes, a robust backend infrastructure handles requests, processes data, and interfaces with AI models—ensuring that generated content is safe, reliable, and accurate, while protecting user privacy and personalization features. Examples of backend infrastructure components for serving LLM agents or interacting with AI models include

- *Caching layers*—Using Redis or Memcached to store frequently accessed data or model responses, the caching layers reduce latency and computational load on the AI models.
- *Containerization and orchestration*—Deploying AI models and associated services involves using Docker containers, managed by orchestration tools like Kubernetes for scalability and ease of management.
- *Serverless functions*—Services like AWS Lambda or Google Cloud Functions can be used to run specific AI-related tasks (like calling APIs) without managing server infrastructure.
- *Model-serving frameworks*—For advanced scenarios, employing specialized tools such as TensorFlow Serving, NVIDIA Triton Inference Server, and Vertex AI can be efficient.

This infrastructure ensures the smooth functioning of the application and efficient communication with external services and APIs.

Data processing

A data processing pipeline is responsible for preprocessing raw input data, feeding it into the AI models, and postprocessing the generated output. This pipeline may